

Fermi–Amaldi-Based Local Mixing Function

Ivan P. Bosko and Viktor N. Staroverov

Department of Chemistry, The University of Western Ontario, London, Ontario N6A 5B7, Canada

(Dated: March 6, 2026)

I. MAIN TEXT

We propose a new local mixing function (LMF) for local hybrid functionals. This LMF is the ratio between the Fermi–Amaldi and the exact exchange energy densities:

$$\text{LMF} = \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}}. \quad (1)$$

The (closed-shell) exact exchange energy density in terms of a one-electron reduced density matrix, $\gamma(\mathbf{r}; \mathbf{r}') = 2 \sum_{i=1}^{N/2} \varphi_i^*(\mathbf{r}) \varphi_i(\mathbf{r}')$, is given by [1, 2]

$$e_x^{\text{exact}}(\mathbf{r}) = -\frac{1}{4} \int \frac{|\gamma(\mathbf{r}; \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \quad (2)$$

The Fermi–Amaldi [3, 4] energy density is expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N ,

$$e_x^{\text{FA}}(\mathbf{r}) = -\frac{1}{2N} \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \quad (3)$$

Both exchange densities are manifestly non-positive and equivalent for one-electron systems and two-electron singlets. Although, for $N > 2$, one can show that

$$e_x^{\text{exact}} \leq e_x^{\text{FA}} \leq 0, \quad (4)$$

since e_x^{FA} is scaled down by N . Combining these properties, we find that the proposed LMF is bound between zero and one:

$$0 \leq \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} \leq 1. \quad (5)$$

This inequality ensures that one hundred percent of exact exchange is employed for one-electron systems, satisfying the one-electron constraint [5, 6].

A particular case of a one-electron limit, i.e., the asymptotic limit ($r \rightarrow \infty$) where the exact exchange must cancel out an excessive self-repulsion of the Coulomb potential, is also satisfied:

$$\lim_{r \rightarrow \infty} \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} = 1 \quad (6)$$

since both energy densities approach $-1/2r$ as $r \rightarrow \infty$.

Another important constraint, the high-density limit,

-
- [1] V. N. Staroverov, Density-functional approximations for exchange and correlation, in *A Matter of Density* (John Wiley & Sons, Ltd, 2012) Chap. 6, pp. 125–156.
 - [2] R. Parr and Y. Weitao, *Density-Functional Theory of Atoms and Molecules*, International Series of Monographs on Chemistry (Oxford University Press, 1989).
 - [3] E. Fermi and E. Amaldi, Le orbite ∞ s degli elementi, *Mem. Reale Accad. Italia* **6**, 117 (1934).
 - [4] P. Bénilan, J. A. Goldstein, and G. R. Rieder, The fermi-amaldi correction in spin polarized thomas-fermi theory, in *Differential Equations and Mathematical Physics*, Mathematics in Science and Engineering, Vol. 186 (Elsevier, 1992) pp. 25–37.
 - [5] J. Jaramillo, G. E. Scuseria, and M. Ernzerhof, Local hybrid functionals, *The Journal of chemical physics* **118**, 1068 (2003).
 - [6] T. M. Maier, A. V. Arbuznikov, and M. Kaupp, Local hybrid functionals: Theory, implementation, and performance of an emerging new tool in quantum chemistry and beyond, *Wiley Interdisciplinary Reviews: Computational Molecular Science* **9**, e1378 (2019).